

Dept. for Speech, Music and Hearing
**Quarterly Progress and
Status Report**

**Automatic matching of vowel
spectra by means of
computers**

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IV. THESIS PROJECT COMPLETED DURING THE PERIOD

A. AUTOMATIC MATCHING OF VOWEL SPECTRA BY MEANS OF COMPUTERS

This thesis work was intended to provide us with some preliminary experience of methods of spectrum matching of vowels by means of computers. One part comprises a study of a simple program for calculating the root mean square difference of the spectrum of a human sample and a synthetic version based on preselected parameter values. The second part comprises a system whereby the machine, given the first guess of formant frequencies calculated the best set of pole frequencies and bandwidths for optimal fit to the human sample.

The strategy of the automatic program is in short as follows. The system function describing the spectrum envelope of the synthetic version is expanded in a Taylor series leading to the specification of linear relations between the errors in all pole frequency and bandwidth data and the differential effects of these errors on any part of the spectrum envelope. The resulting set of linear equations is expressed in an orthogonalized matrix form and solved for optimal values of synthesis parameters. This method has the drawback that gross errors in the preselected estimate of parameter data often lead to unnatural solutions. The time necessary for a matching was 3 minutes.

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